# Average Rate of Change

**ANSWER KEY**

## Section 1: Formula and Secant Line Concept

**1** State the formula for the average rate of change of $f(x)$ over $[a, b]$.

**a.** $\frac{f(b)-f(a)}{b-a}$ ✓

**b.** $\frac{f(a)+f(b)}{2}$

**c.** $\frac{f(b)-f(a)}{2}$

**d.** $\frac{b-a}{f(b)-f(a)}$

The average rate of change is the slope of the secant line passing through $(a, f(a))$ and $(b, f(b))$: $\frac{f(b)-f(a)}{b-a}$.

**2** Find the average rate of change of $f(x) = x^2 - 3x$ from $x = 1$ to $x = 4$.

**a.** 2 ✓

**b.** 4

**c.** 3

**d.** 1

$f(1) = 1 - 3 = -2$. $f(4) = 16 - 12 = 4$. Rate $= \frac{4-(-2)}{4-1} = \frac{6}{3} = 2$.

**3** Find the slope of the secant line connecting $(2, f(2))$ and $(5, f(5))$ for $f(x) = 2x^2 + 1$.

**a.** 14 ✓

**b.** 12

**c.** 16

**d.** 10

$f(2) = 9$, $f(5) = 51$. Slope $= \frac{51-9}{5-2} = \frac{42}{3} = 14$.

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## Section 2: Tabular and Data-Based Rates of Change

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**4** A table shows $t = [0, 2, 5]$ and distance $s(t) = [0, 24, 75]$. What is the average velocity from $t = 2$ to $t = 5$?

**a.** $17$ m/s ✓

**b.** $15$ m/s

**c.** $25$ m/s

**d.** $12$ m/s

Average velocity $= \frac{s(5) - s(2)}{5 - 2} = \frac{75 - 24}{3} = \frac{51}{3} = 17$ m/s.

**5** Water depth in a reservoir drops from $80$ ft at day $4$ to $68$ ft at day $10$. Find the average rate of change in depth per day.

**a.** $-2$ ft/day ✓

**b.** $2$ ft/day

**c.** $-1.2$ ft/day

**d.** $-12$ ft/day

$\frac{68 - 80}{10 - 4} = \frac{-12}{6} = -2$ ft/day.

**6** If the average rate of change of $g(x)$ over $[1, 6]$ is $4$ and $g(1) = 5$, find $g(6)$.

**a.** $25$ ✓

**b.** $20$

**c.** $29$

**d.** $24$

$\frac{g(6) - 5}{6 - 1} = 4 \Rightarrow g(6) - 5 = 20 \Rightarrow g(6) = 25$.

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## Section 3: Algebraic Expressions with Variables

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**7** Find the average rate of change of $f(x) = x^2$ over the interval $[2, 2+h]$ in terms of $h$.

**a.** $4 + h$ ✓

**b.** $4$

**c.** $2 + h$

**d.** $4 + 2h$

$\frac{f(2+h)-f(2)}{h} = \frac{(2+h)^2-4}{h} = \frac{4+4h+h^2-4}{h} = \frac{4h+h^2}{h} = 4+h.$

**8** Find the average rate of change of $f(x) = 3x^2 - 5$ over $[x, x+h]$.

**a.** $6x + 3h$ ✓

**b.** $6x$

**c.** $3x + 3h$

**d.** $6x - 5$

$\frac{3(x+h)^2-5-(3x^2-5)}{h} = \frac{6xh+3h^2}{h} = 6x+3h.$

**9** For $f(x) = \frac{1}{x}$, find the average rate of change over $[a, b]$.

**a.** $-\frac{1}{ab}$ ✓

**b.** $\frac{1}{ab}$

**c.** $-\frac{1}{a+b}$

**d.** $\frac{a-b}{ab}$

$\frac{1/b-1/a}{b-a} = \frac{(a-b)/(ab)}{b-a} = -\frac{b-a}{ab(b-a)} = -\frac{1}{ab}.$

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## Section 4: Contextual Applications

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**10** A projectile's height is given by $h(t) = -16t^2 + 80t$. Find its average velocity over the first 3 seconds $[0, 3]$.

**a.** 32 ft/s ✓

**b.** 48 ft/s

**c.** 16 ft/s

**d.** 80 ft/s

$h(0) = 0$. $h(3) = -16(9) + 80(3) = -144 + 240 = 96$ ft. Average velocity $= \frac{96 - 0}{3 - 0} = 32$ ft/s.

**11** A company's revenue in thousands of dollars is $R(x) = 50x - x^2$, where $x$ is in hundreds of units. What is the average rate of change in revenue as production increases from $x = 10$ to $x = 20$?

**a.** 20 thousand dollars per unit ✓

**b.** 30

**c.** 15

**d.** 25

$R(10) = 500 - 100 = 400$. $R(20) = 1000 - 400 = 600$. Rate $= \frac{600 - 400}{20 - 10} = \frac{200}{10} = 20$.

**12** Why does a linear function $f(x) = mx + b$ have the same average rate of change over EVERY interval?

**a.** **Because the slope of the secant line between any two points is constantly** $m$ ✓

**b.** Because $b = 0$

**c.** Because the function has no intercepts

**d.** Because $f(x)$ is symmetric

For any $a$ and $b$, $\frac{(mb + b_0) - (ma + b_0)}{b - a} = \frac{m(b - a)}{b - a} = m$, which is constant.

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